

## Introduction

Capacitor inductance measurement is in general complex due to the fact that the parasitic  $L_s$  value is very low (nH). In general it is possible to measure it with a component analyzer, but it is mandatory to compensate properly the wires (short/open test) before the measurement and the complete system (capacitor + wires) should not be moved during all the test.

When low capacitor inductance values have to be measured or when capacitor has more than 1 pair of terminals (external busbar necessary for the measurement), it is strongly recommended to use low voltage discharge method, at least to confirm the value obtained with the component analyzer.

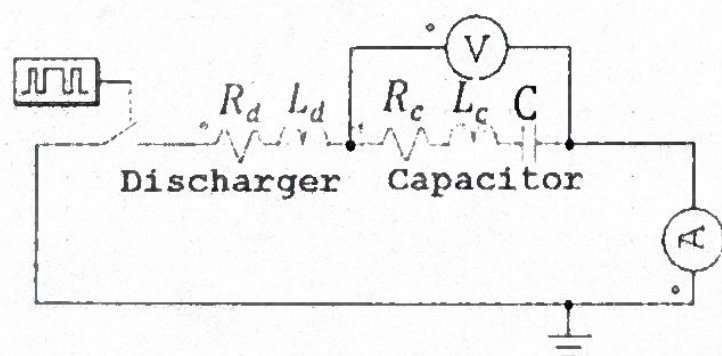
## Theory

Low voltage surge discharge inductance measurement method is based on the fact that a capacitor is always discharged at the resonance frequency of the loop.

Discharge has to be provoked short-circuiting its terminals with the minimum possible impedance (inductance and resistance). Inductance of the external element used for the short-circuit is subtracted from the total inductance of the loop based on the voltage waveform.

- In case the capacitor has only two terminals, it can be discharged directly with a cooper plate (as short as possible and with good section). Example: MKP capacitors.
- In case the capacitor has more than 1 terminal per polarity and terminals with same polarity are going to be short-circuited in the application with a busbar, it would be necessary to use an overlapped external busbar to connect them and assure that the inductance from the point where the short-circuit is provoked to all the pairs of terminals is the same. Example: most MKK capacitors.
- In case the capacitor has an integrated busbar and has been designed to connect directly to the semiconductors, inductance has to be measured at each pair of terminals. Example: PCC-LP

Below in red we have the capacitor represented as its equivalent circuit: capacity, ESR and ESL.  
In green we have the element used to short-circuit the capacitor.



Following Kirchhoff's law and solving the differential equation of the RLC circuit considering low damping coefficient, current and voltage equation are obtained. Dividing current equation for two consecutive maximum, Equation 1 is obtained. Equation 2 is the damping coefficient. Equation 3 represents the natural frequency as function of resonance frequency and damping coefficient, Equation 4 shows the natural frequency and finally Equation 5 distinguishes between the inductance of the capacitor and the total loop inductance (including the short-circuiting element):

$$e^{aT} = \frac{I_{p1}}{I_{p2}} \quad (1)$$

$$a = \frac{R}{2L} \quad (2)$$

where:

$$R = R_d + R_c$$

$$L = L_d + L_c$$



$$\omega_0 = \sqrt{\omega_r^2 + \alpha^2} \quad (3)$$

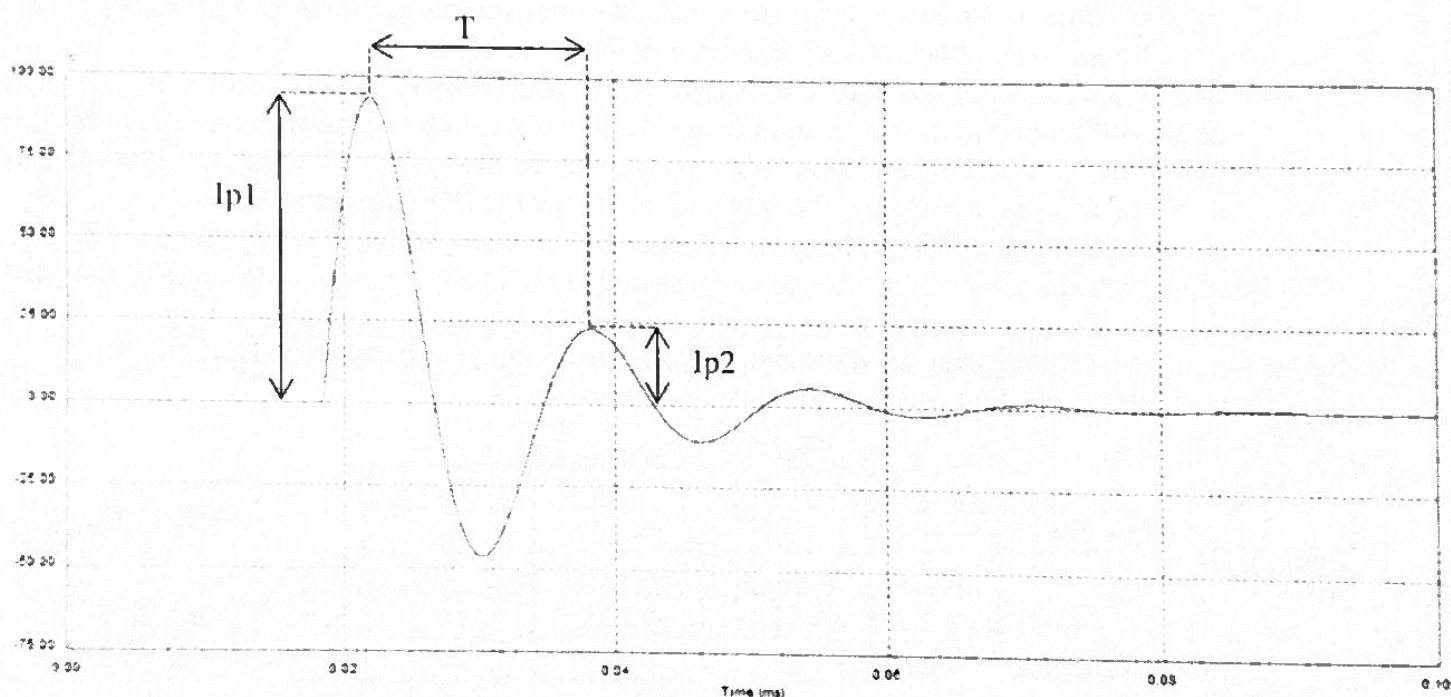
$$\omega_0 = 1 / \sqrt{LC} \quad (4)$$

$$\frac{V_1}{V_c} = \frac{L_c}{L} \quad (5)$$

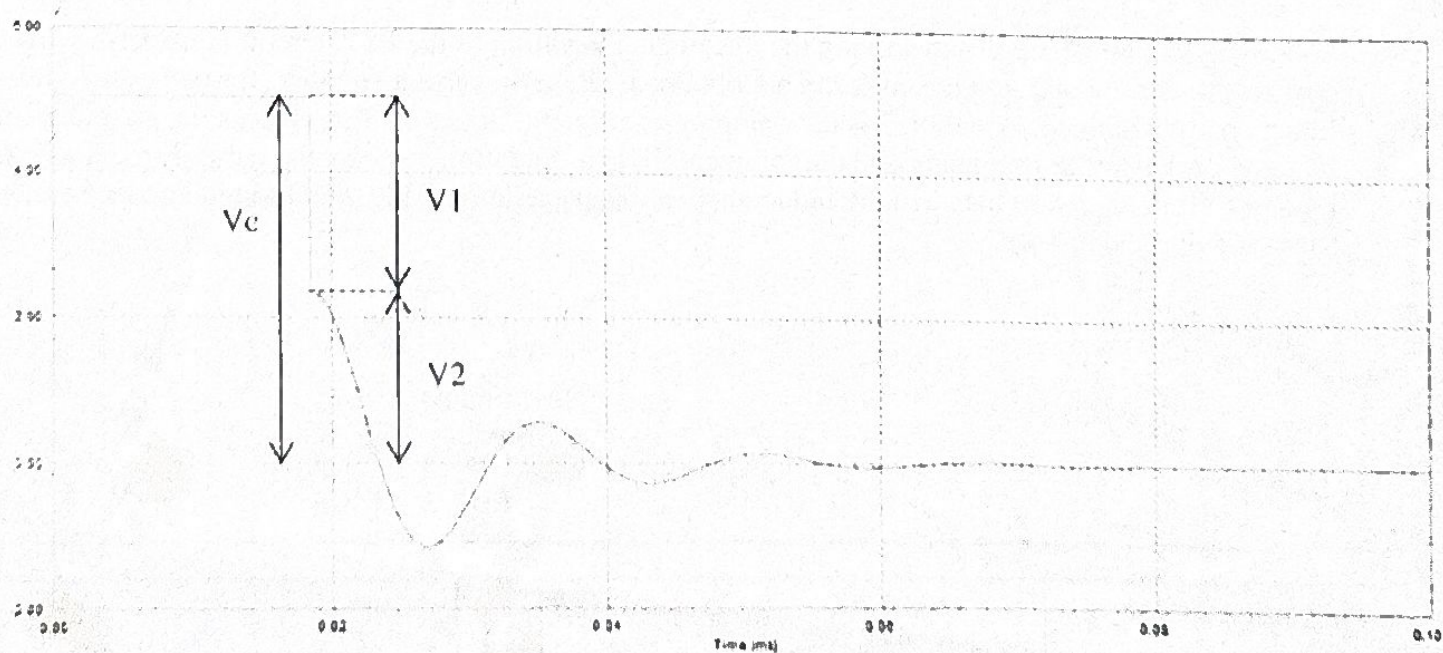
$$\omega_r = \frac{2\pi}{T}$$

Below current and voltage waveforms and main parameters necessary for the inductance calculation represented:

Discharge current waveform



Discharge voltage waveform



Procedure

Although the voltages used for this test are very low (<20V), due to the low inductance, the peak of current is normally high, especially when high capacitance/low inductance capacitors are measured (up to some kA). In a first approach this peak of current can be estimated by the formula below:

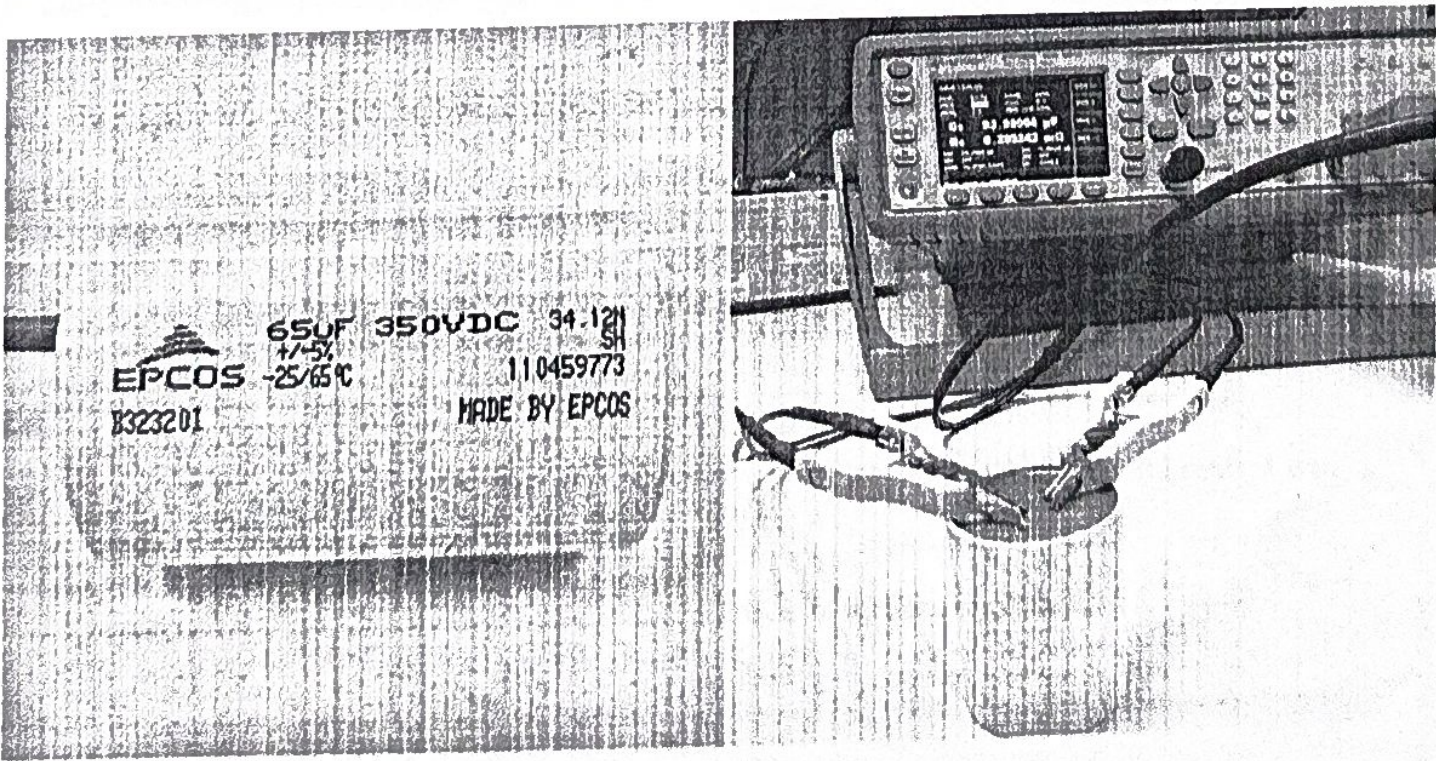
I<sub>p1</sub> = V<sub>c</sub> √(C/L)

As general rule V<sub>c</sub> has to be as low as possible to have clear current and voltage waveforms. Recommended voltage values above 4V and always below 20V. It is important to use the appropriate current and voltage sensors, with amplitude ranges according to the expected values. For current monitoring use a Rogowski coil. Both sensors have to be connected to the same oscilloscope. Use voltage drop as triggering factor. Oscilloscope time scale adjustment has to be carried out according to the formula below, where T is the expected period of the waveform (in the reality it will be slightly higher due to damping effect).

T = 2π√LC

Example

This is the description of the test, step by step.  
AC capacitor is used as example:



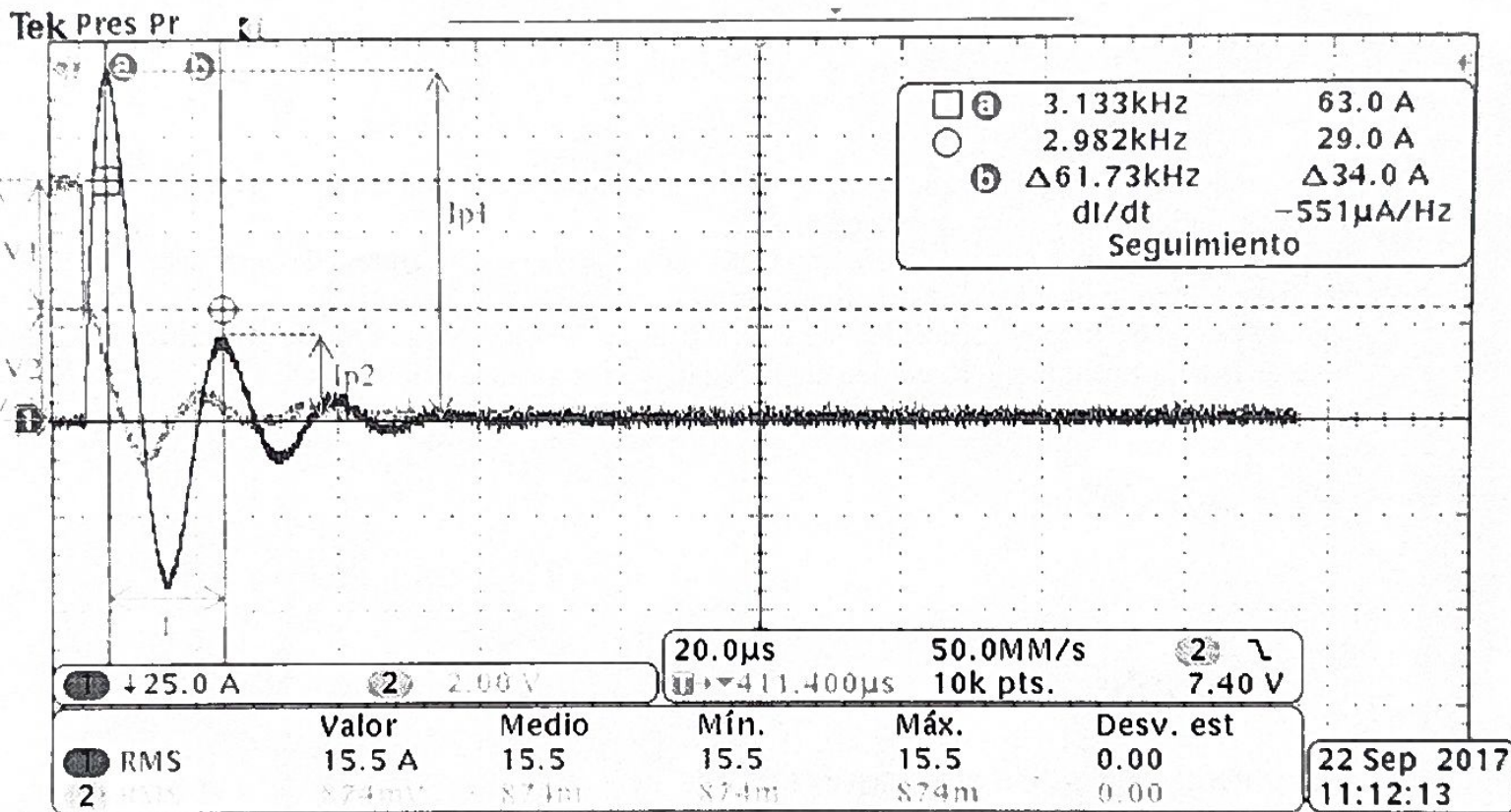
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# Capacitor inductance measurement based on low voltage surge discharge method

Below the current (dark blue) and voltage (light blue) waveforms obtained during the discharge with all the values necessary to carry out the inductance calculation remarked:



$I_{p1} = 92A$   
 $I_{p2} = 22.5A$   
 $T = 16.2\mu sec$   
 $V_c = 5.04V$   
 $V_1 = 2.72V$   
 $V_2 = 2.32V$   
 $C = 63.3\mu F$  (real value measured)

Now that all these values are defined, applying the formulas in the order listed below, capacitor inductance value will be calculated:

## Step 1: damping coefficient

$$e^{aT} = \frac{I_{p1}}{I_{p2}} \longrightarrow a = 87097 = \frac{R}{2L}$$

## Step 2: resonance and natural frequencies

$$\omega_r = \frac{2\pi}{T} \longrightarrow \omega_r = 387664 \text{ rad/sec} \longrightarrow f_r = 61730 \text{ Hz}$$

$$\omega_0 = \sqrt{\omega_r^2 + a^2} \longrightarrow \omega_0 = 397327 \text{ rad/sec}$$

As in general film capacitors have a low  $R_s$  value (low damping coefficient), in most of the cases resonance and natural frequencies are very similar.

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Step 3: loop inductance calculation (L)

$\omega_0 = 1 / \sqrt{LC} \longrightarrow L = 100.1 \text{ nH}$

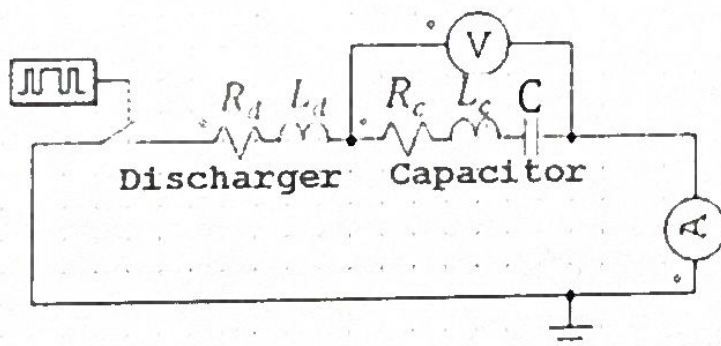
Step 4: capacitor inductance

$\frac{V_1}{V_c} = \frac{L_c}{L} \longrightarrow L_c = 54 \text{ nH}$

So capacitor inductance is 54nH.

Check the result

One of the advantages of this method is that after the test, there is always the chance to check that the result is correct using any simulation software and comparing the obtained waveforms and the real ones. We strongly recommend carrying out this step in order to probe that validity of the results. Circuit is the same showed in the first page. Total resistance is extracted from the damping formula.  $R_d$  is the ESR at the frequency of the discharge: in this case 61730 Hz.  $R_d$  is obtained subtracting  $R - R_c$ .

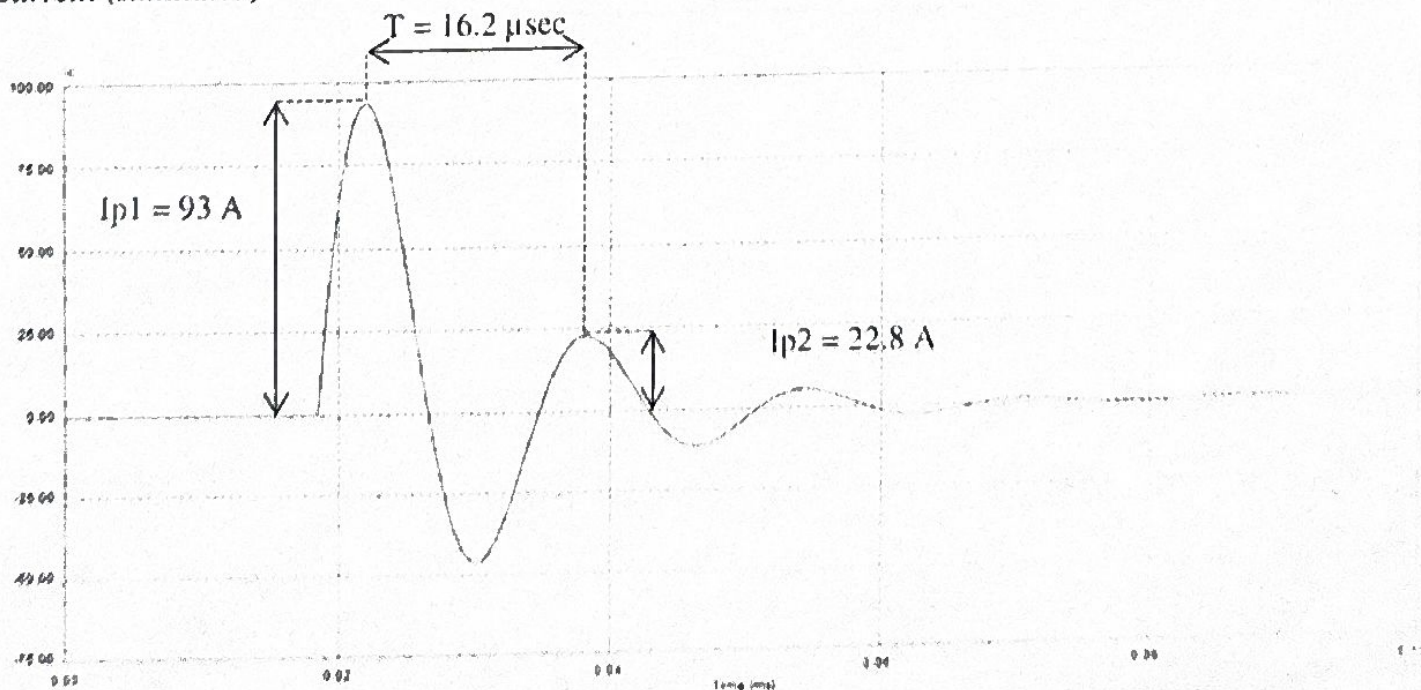


$a = \frac{R}{2L} \longrightarrow R = 17.4 \text{ mohm}$   
 $R_c = 9.78 \text{ mohm}$   
 $R_d = 7.62 \text{ mohm}$

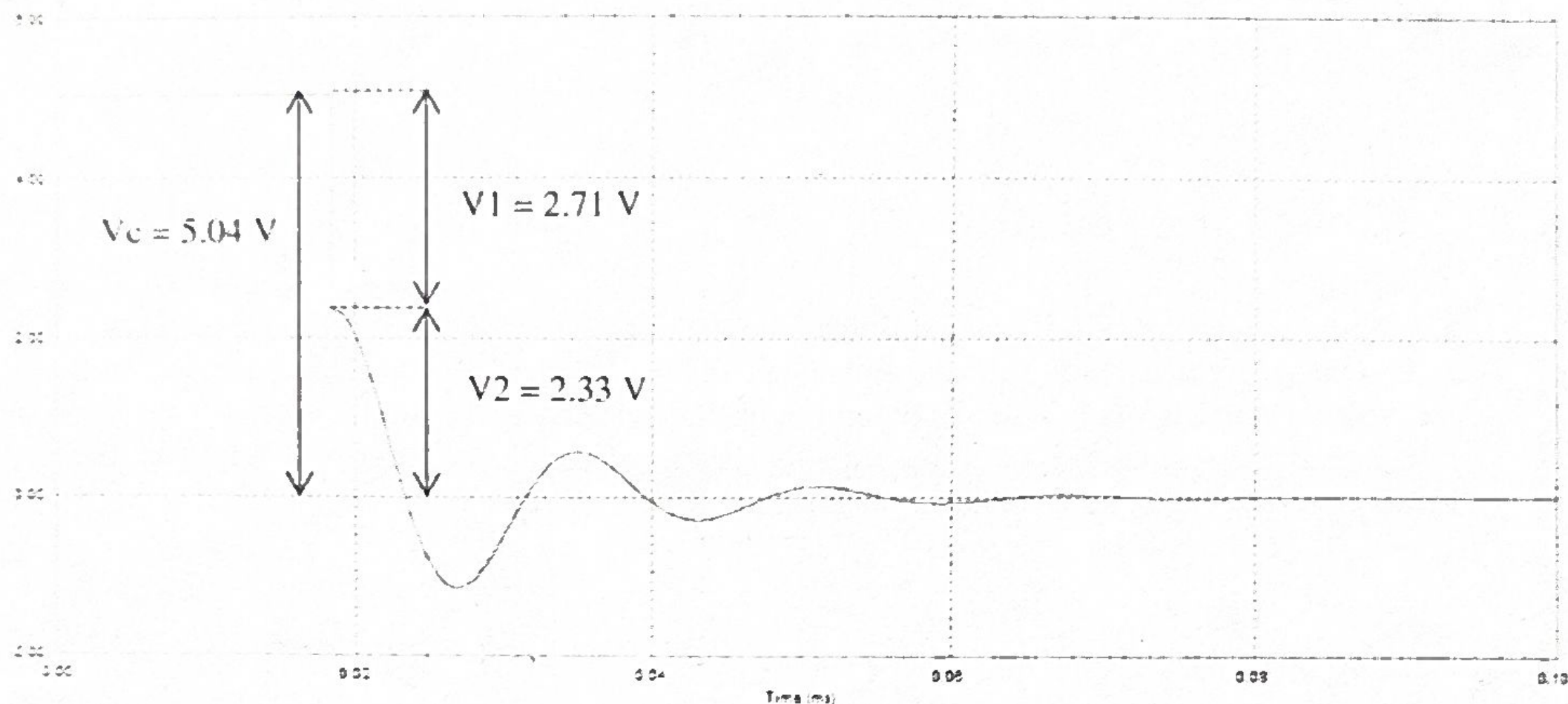
where:  
 $R = R_d + R_c$   
 $L = L_d + L_c$

If the circuit above is simulated substituting all the values obtained during the calculation, the waveforms obtained are:

Current (simulated)

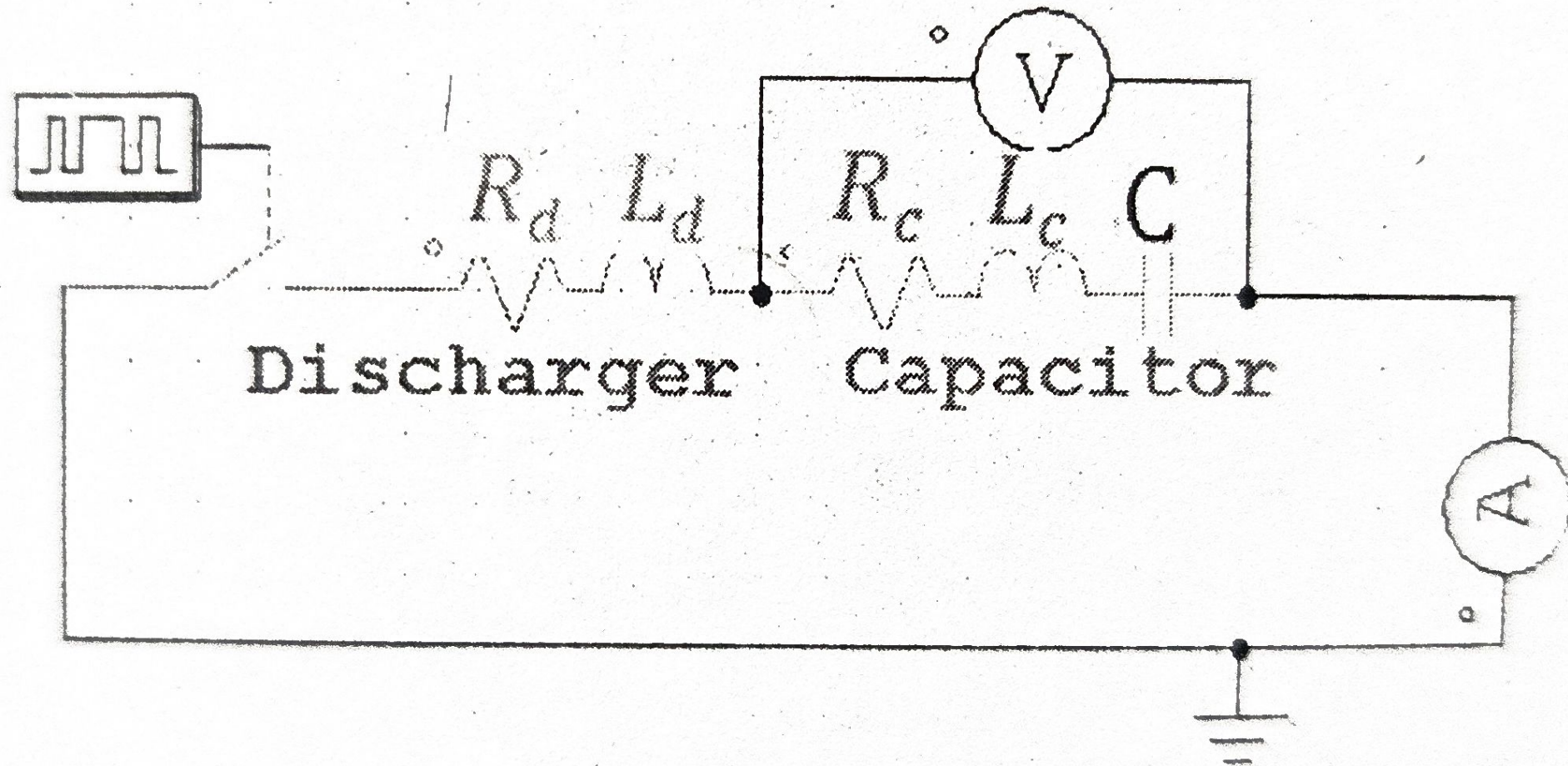


Voltage(simulated)

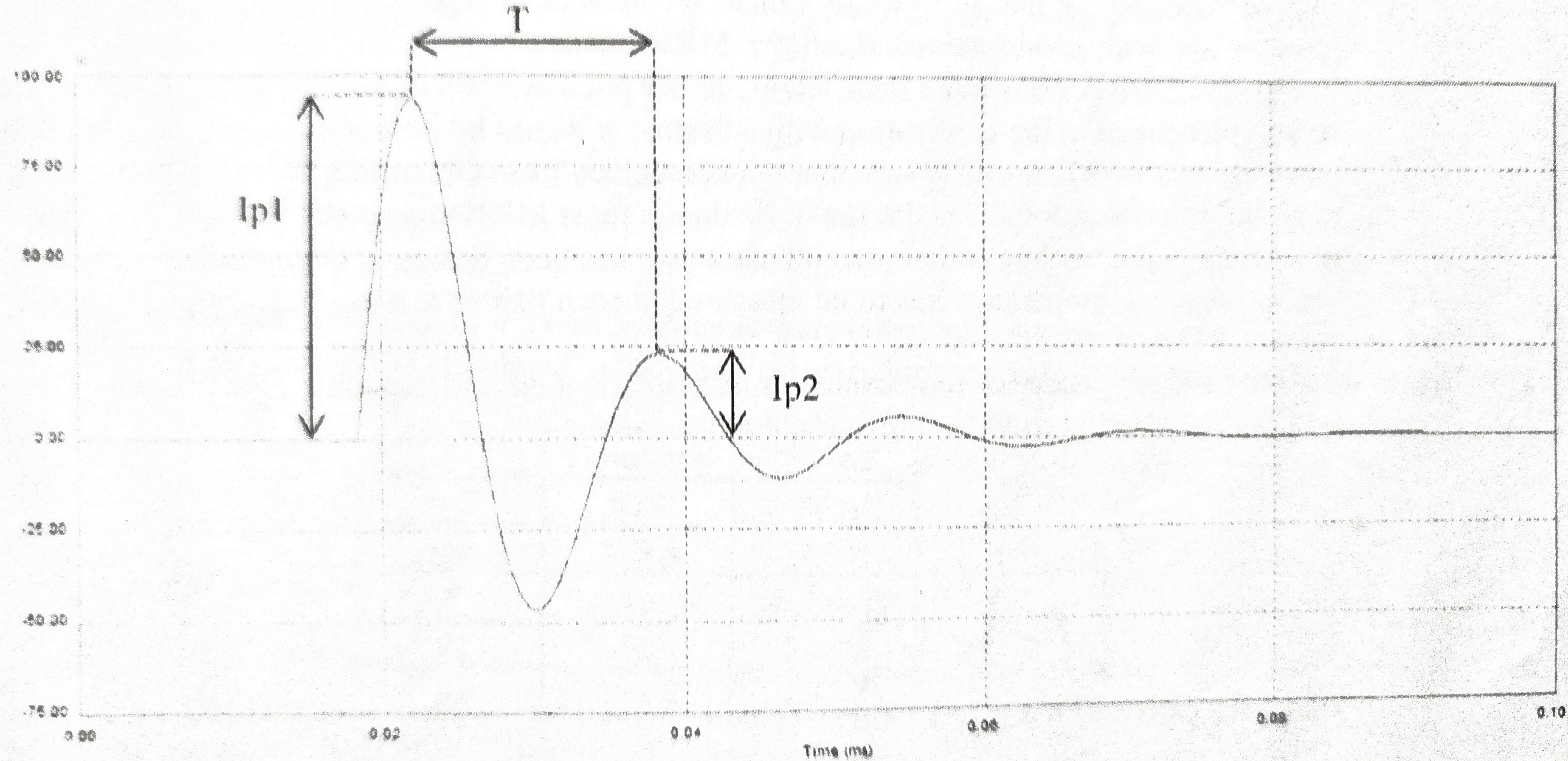


If these waveforms are compared to the oscilloscope real captures, both are equal, so the calculation has been carried out correctly. Below the simulated values and real ones for the parameters involved in the calculation:

|     | Simulated            | Tested               |
|-----|----------------------|----------------------|
| Ip1 | 93 A                 | 92 A                 |
| Ip2 | 22,8 A               | 22,5 A               |
| T   | 16,2 $\mu\text{sec}$ | 16,2 $\mu\text{sec}$ |
| V1  | 2,71 V               | 2,72 V               |
| V2  | 2,33 V               | 2,32 V               |

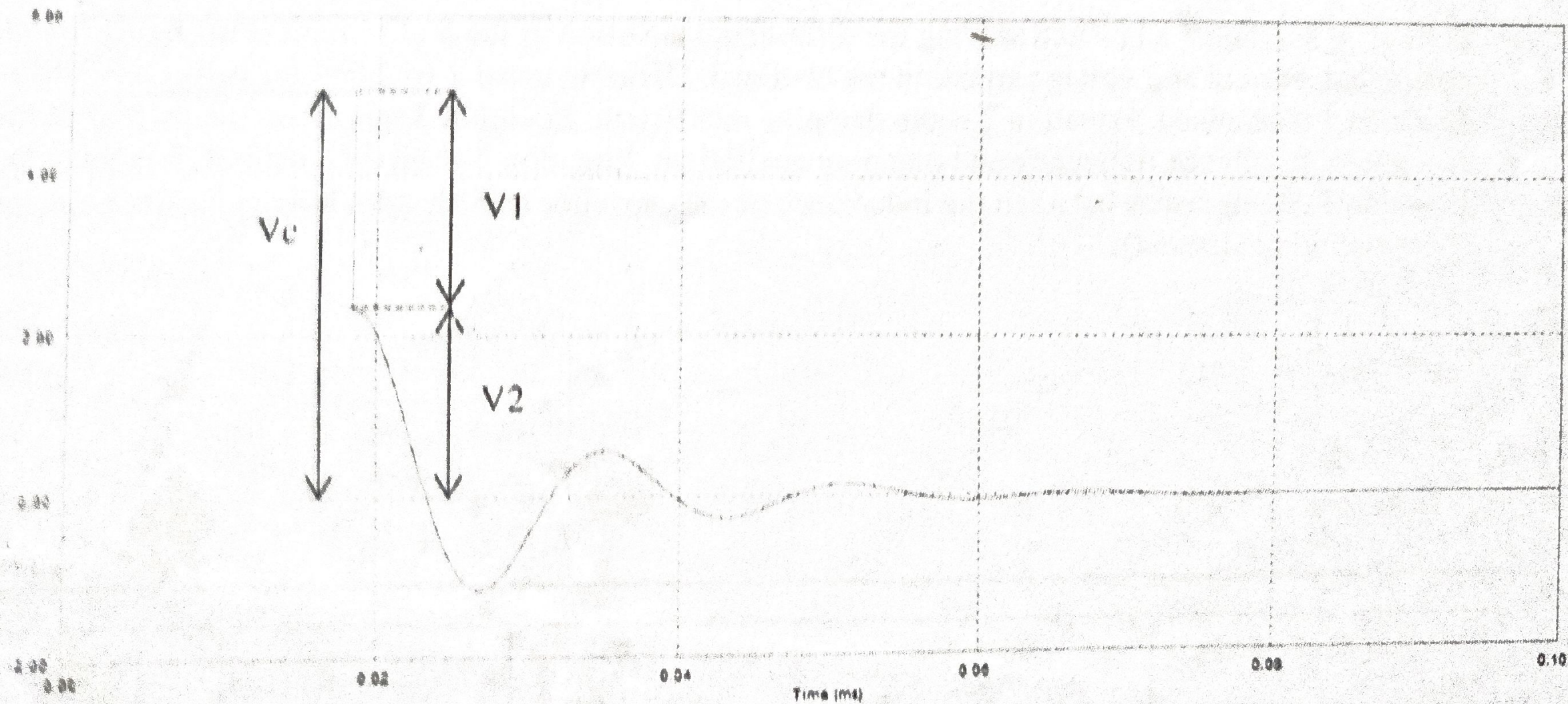


# Discharge current waveform

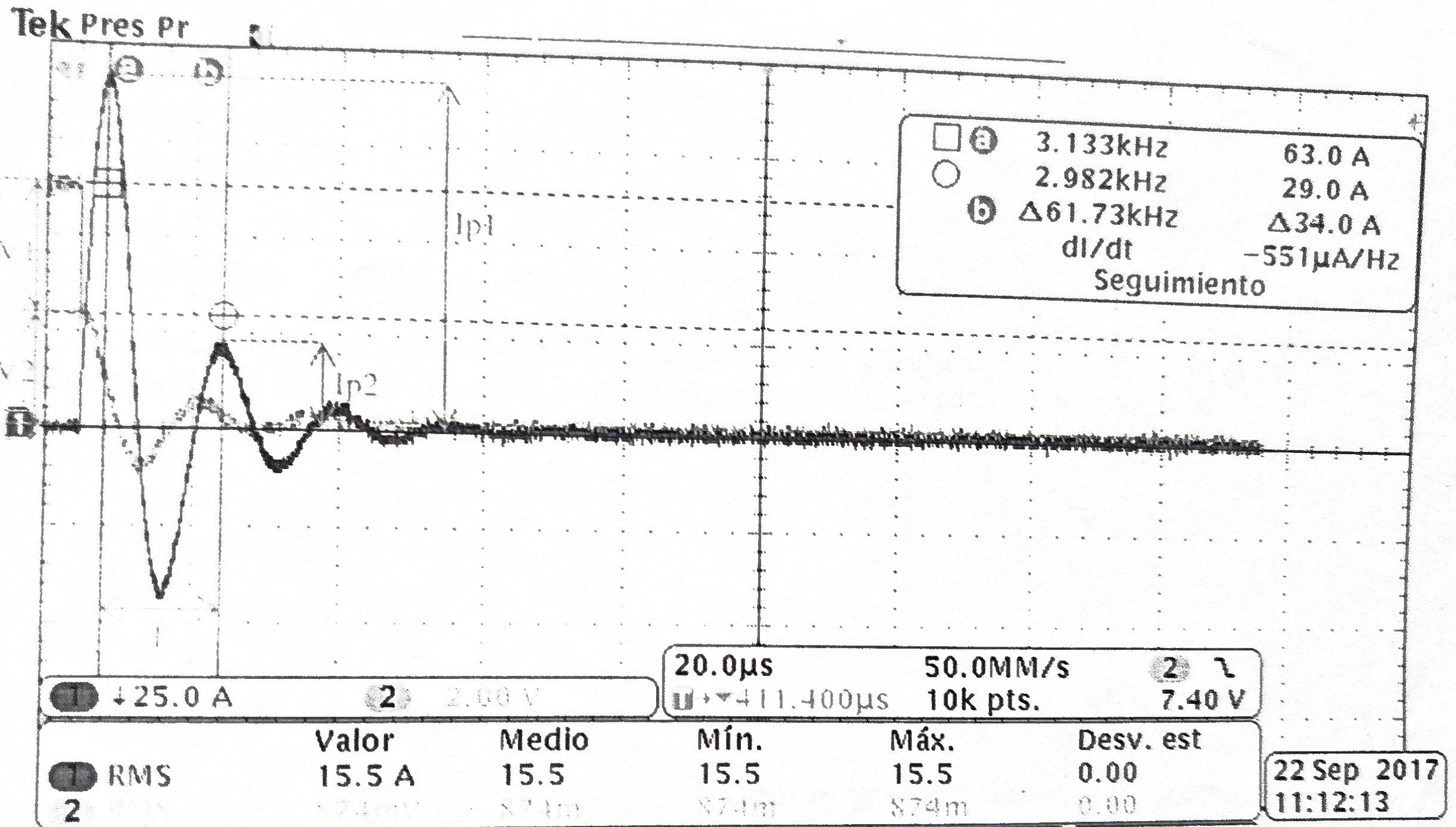


or registration of a utility model or design.

# Discharge voltage waveform



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